Does Domain-Specific Embedding Fine-Tuning Improve RAG Performance on Financial Document QA? A Multi-Model Study

Anvi Kudaraya
University of California, Merced
akudaraya@ucmerced.edu
https://github.com/anviik/RAG-Comparison/tree/main

Abstract

Retrieval-Augmented Generation (RAG) has become a popular way to ground LLM outputs in actual document content, but it is not well understood how much domain-specific embedding fine-tuning actually helps downstream QA quality, especially across different model tiers. We evaluate three LLMs, GPT-4o-mini (OpenAI, 2023), Claude-3.5-Haiku (Anthropic, 2024), and Mistral-7B (Jiang et al., 2023), on 150 questions from ten 2022 annual report filings using the FinanceBench benchmark (Islam et al., 2023). We test combinations of base versus fine-tuned embeddings (Reimers and Gurevych, 2019), two temperatures (0.0 and 0.7), and two retrieval depths ($k \in \{3, 5\}$). Fine-tuned embeddings improve ROUGE-1 modestly (+0.002 to +0.004), with the biggest LLM-as-judge gain going to Mistral-7B (+0.060). That said, RAG itself drives most of the quality improvement: Claude-3.5-Haiku's judge score jumps +0.83 just from adding retrieval. Numeric accuracy is low across the board (at most 21.6%) and actually drops with fine-tuning, which was not what we expected. Overall, retrieval quality is the main bottleneck, and numeric QA remains a hard unsolved problem even for strong commercial models.

1 Introduction

Question answering over financial documents has real applications in investment analysis and regulatory compliance, but 10-K filings are dense and highly structured, with tables, footnotes, and domain-specific language that trips up even capable models (Chen et al., 2021). LLMs are prone to producing plausible-sounding but factually wrong answers when asked about specific documents (Shuster et al., 2021).

RAG addresses this by retrieving relevant chunks at inference time (Lewis et al., 2020), but financial documents are particularly tricky: numbers appear in multiple contexts and critical information is often locked in tables that don't chunk cleanly. Prior work showed these retrieval challenges cause failures even for GPT-4 (Islam et al., 2023). One underexamined piece of the pipeline is the embedding model itself, since there is evidence that domain fine-tuning improves retrieval on specialized tasks (Thakur et al., 2021).

We ask whether fine-tuning retrieval embeddings on financial text improves RAG performance, and whether the effect varies across model capability tiers. Our hypothesis was that stronger commercial models can partially compensate for weaker retrieval using in-context reasoning, while smaller open-source models depend more heavily on what the retriever returns (Mallen et al., 2023). Our main contributions are: (1) a systematic comparison of base versus fine-tuned embeddings across three model tiers; (2) analysis of how temperature, retrieval depth, and context window limits interact; and (3) a finding that embedding fine-tuning improves fluency metrics but consistently hurts numeric accuracy.

2 Related Work

2.1 LLMs in Financial Data

Financial NLP spans early sentiment analysis to domain-specific pretraining. Yang et al. (2020) showed pretraining BERT on financial text improved classification, but QA over structured documents is harder. Chen et al. (2021) found even large models struggled on multi-step numerical reasoning, and Zhu et al. (2021) studied table-and-text hybrid QA directly relevant to 10-K filings. FinanceBench (Islam et al., 2023) confirmed GPT-4 still fails on exact numerical extraction. Our work differs from domain-specific approaches like BloombergGPT (Wu et al., 2023) by targeting retrieval quality through embedding fine-tuning rather than full model training.

2.2 Retrieval-Augmented Generation

Lewis et al. (2020) introduced RAG to combine parametric knowledge with retrieved context. DPR (Karpukhin et al., 2020) showed dense retrieval outperforms BM25, and REALM (Guu et al., 2020) integrated retrieval into pretraining. Retrieval quality matters critically: irrelevant passages actively degrade LLM performance (Shi et al., 2023). We use FAISS (Johnson et al., 2019) for similarity search and Sentence-BERT (Reimers and Gurevych, 2019) as our base embedding model. While Ni et al. (2022) showed scaling general-purpose embeddings helps cross-domain retrieval, our low-resource domain adaptation is motivated by BEIR (Thakur et al., 2021), which found general-purpose retrievers degrade significantly on out-of-domain tasks.

3 Method

Our pipeline follows the standard RAG setup (Lewis et al., 2020). Each 10-K PDF is processed with PyMuPDF, chunked into 500-token segments with 50-token overlap, embedded, and indexed into a FAISS vector store (Johnson et al., 2019). At query time, the top-k chunks are retrieved by cosine similarity and prepended to the query as context. The LLM is prompted to answer based only on the provided context, with no few-shot examples.

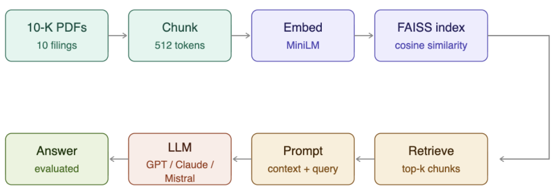

Figure 1: Overview of the RAG pipeline. Annual report PDFs are parsed and chunked, embedded into a FAISS vector store, retrieved at query time, and passed to the LLM as grounded context for financial question answering.

Embedding Fine-Tuning. We fine-tune `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019) on 35 financial question-passage pairs using MultipleNegativesRankingLoss for 10 epochs. The training set is intentionally small to simulate low-resource domain adaptation in a realistic setting. The fine-tuned model is only evaluated on held-out questions.

4 Experiments

4.1 Setup

4.1.1 Benchmark and Dataset

We build on FinanceBench (Islam et al., 2023) using ten 2022 10-K filings: Coca-Cola, Amazon, American Express, Apple, Best Buy, eBay, JPMorgan Chase, Netflix, Oracle, and Walmart. For each company we wrote 15 questions across three types: **factual** (e.g., "What was total revenue in FY2022?"), **numerical** (e.g., "By what percentage did operating income change YoY?"), and **reasoning** (e.g., "What risks does the company's debt structure present?"). This gives 150 questions total (50 per type), each with a gold-standard reference answer.

4.1.2 Models

GPT-4o-mini (OpenAI, 2023): top-tier commercial API benchmark. **Claude-3.5-Haiku** (Anthropic, 2024): commercial API model optimized for structured tasks. **Mistral-7B** (Jiang et al., 2023): open-source model run locally in 4-bit quantization via llama.cpp on a Google Colab T4 GPU, with a 4,096-token context window limit.

4.1.3 Implementation Details

We test all combinations of embedding type (base, fine-tuned), temperature (0.0, 0.7), and $k \in \{3, 5\}$, plus a no-RAG baseline for GPT-4o-mini and Claude-3.5-Haiku, 3,900 total evaluated responses. Metrics: **ROUGE-1/L** (Lin, 2004) (token overlap); **Exact Match** (verbatim string; 0.0 for all conditions); **LLM-as-judge** (0–3 scale rated by GPT-4o-mini (Zheng et al., 2023)); **Numeric accuracy** ($\pm 10\%$ tolerance); **Latency** (wall-clock seconds; Mistral reflects local GPU inference, GPT/Claude reflect API round-trip).

4.2 Quantitative Results

Table 1 reports results at our primary setting (temp=0.0, k=5). GPT-4o-mini achieves the highest ROUGE-1 (0.118 fine-tuned), but Claude-3.5-Haiku leads on LLM Judge (2.067) despite ranking last on ROUGE-1. This inversion reflects how ROUGE penalizes Claude's verbose but semantically accurate responses against short reference answers. We treat the LLM Judge as the more meaningful signal. Mistral-7B scores substantially lower across the board (best judge: 1.307), expected given its smaller size, quantization, and context window constraints.

Model	Embedding	ROUGE-1	ROUGE-L	LLM Judge (0–3)	Numeric Acc.	Latency (s)
GPT-4o-mini	Base	0.115	0.107	1.933	21.6%	2.20
GPT-4o-mini	Fine-tuned	0.118	0.108	1.940	13.5%	2.78
Claude-3.5-Haiku	Base	0.054	0.048	2.067	18.9%	2.92
Claude-3.5-Haiku	Fine-tuned	0.058	0.053	2.067	16.2%	2.90
Mistral-7B	Base	0.076	0.070	1.247	13.5%	4.78
Mistral-7B	Fine-tuned	0.079	0.073	1.307	5.4%	4.85
GPT-4o-mini	No-RAG	0.093	–	1.387	–	–
Claude-3.5-Haiku	No-RAG	0.043	–	1.233	–	–

Table 1: Main results (temp=0.0, k=5). Exact Match was 0.0 for all conditions (omitted). Numeric Acc. = numeric accuracy on numerical questions ($\pm$10% tolerance). Mistral latency reflects local GPU inference; GPT-4o-mini and Claude-3.5-Haiku reflect API round-trip time.

		ROUGE-1			LLM Judge (0–3)		
Model	Embedding	Factual	Numerical	Reasoning	Factual	Numerical	Reasoning
GPT-4o-mini	Base	0.124	0.112	0.097	1.902	1.730	2.258
GPT-4o-mini	Fine-tuned	0.132	0.101	0.099	1.963	1.595	2.290
Claude-3.5-Haiku	Base	0.055	0.039	0.070	2.085	1.838	2.290
Claude-3.5-Haiku	Fine-tuned	0.062	0.037	0.072	2.049	1.757	2.484
Mistral-7B	Base	0.081	0.055	0.089	1.195	0.865	1.839
Mistral-7B	Fine-tuned	0.086	0.048	0.096	1.366	0.730	1.839

Table 2: Results by question type (temp=0.0, k=5). Fine-tuning consistently helps factual and reasoning questions but hurts numerical questions across all models. Numerical questions are the hardest by both metrics.

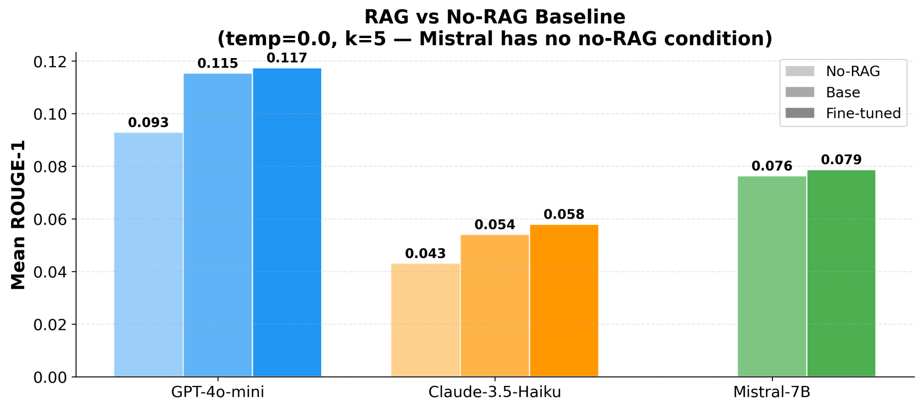

Figure 2: ROUGE-1 under no-RAG, base RAG, and fine-tuned RAG (temp=0.0, k=5). The biggest gain comes from adding retrieval; fine-tuning adds a smaller incremental improvement.

The more important takeaway is how much simply having RAG helps: Claude's judge score jumps from 1.233 (no-RAG) to 2.067 (fine-tuned RAG, +0.834), while fine-tuning on top adds +0.000. For GPT: +0.553 from retrieval, +0.007 from fine-tuning. The bottleneck is retrieval, not the embedding model tier.

Table 2 breaks down performance by question type. Fine-tuning consistently helps factual and reasoning questions but hurts numerical questions across all models (e.g., GPT factual ROUGE-1: +0.008; GPT numerical: -0.010). Numerical is the hardest type overall, with judge scores 0.2–0.5 points below factual for the same model.

4.3 Qualitative Results

4.3.1 ROUGE vs. LLM Judge Inversion

Claude ranks last on ROUGE-1 but first on LLM Judge. When the reference answer is short like "$22.8 billion" and Claude responds with a correct but multi-sentence explanation, ROUGE penalizes the extra length. GPT-4o-mini's terser, more extractive style aligns better with short references. ROUGE alone will systematically underestimate verbose models even when their answers are correct.

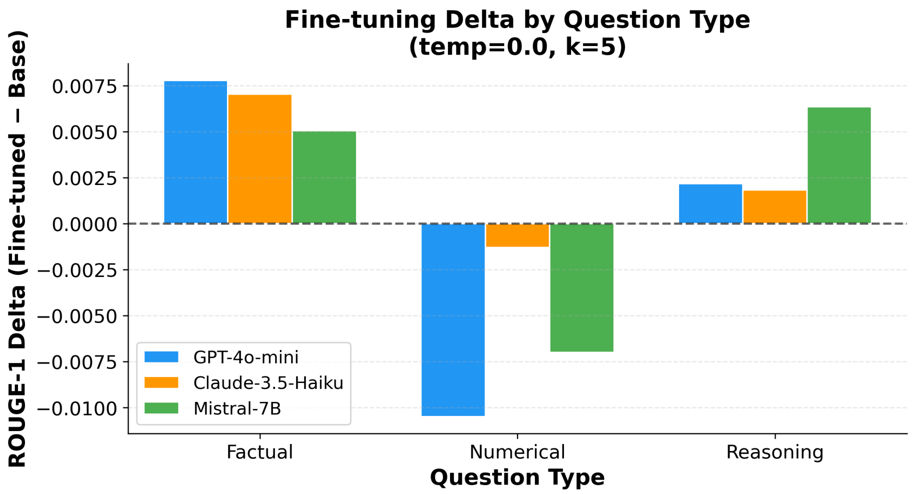

Figure 3: Fine-tuning ROUGE-1 delta (fine-tuned − base) by question type (temp=0.0, k=5). Numerical questions show consistent negative deltas; factual and reasoning questions benefit from fine-tuning.

4.3.2 Company-Level Variation

Figure 4 shows per-company ROUGE-1. Oracle is consistently the hardest across all three models (0.044, 0.043, 0.022), likely because its non-GAAP reconciliations and segment disclosures don't chunk cleanly. Netflix and Walmart are the easiest for GPT (both 0.197). JPMorgan ranks among GPT and Mistral's strongest companies (0.158 and 0.131) but is one of Claude's weakest (0.050), suggesting performance depends on both document structure and model vocabulary.

4.3.3 Numeric Accuracy

Even at its best (GPT-4o-mini base, 21.6%), the majority of numerical questions are answered incorrectly by every model. Fine-tuning makes this worse across the board: GPT drops 8.1 percentage points, Claude 2.7, and Mistral 8.1. We dig into why in the case study below.

4.3.4 Retrieval Confidence vs. Answer Quality

For GPT and Claude, retrieval confidence and judge score are essentially uncorrelated (r=0.03, p >0.5); for Mistral the correlation is moderate and significant (r=0.43, p <0.0001). Commercial models can fall back on parametric knowledge when retrieval is weak (Mallen et al., 2023), while Mistral's answers are more directly tied to what the retriever returns. Fine-tuned embeddings also produce lower average confidence scores than base embeddings (0.530 vs. 0.596) despite equal or better answer quality, so confidence thresholds won't transfer cleanly across embedding setups.

4.4 Ablation Study

4.4.1 Temperature

Temperature has almost no effect on GPT and Claude (<0.002 difference) but noticeably hurts Mistral (−0.006 at temp=0.7), shown in Figure 7. For financial QA this makes intuitive sense: questions like "What was Netflix's net income in 2022?" have a single correct answer, and adding randomness only introduces error. Temp=0.0 is at least as good as 0.7 in every condition we tested.

4.4.2 Retrieval Depth

Table 3 and Figure 8 show the k ablation results. GPT and Claude both improve with k=5 (+0.010 and +0.003 respectively). Mistral is consistently hurt: for fine-tuned embeddings, ROUGE-1 drops from 0.092 at k=3 to 0.079 at k=5 (−0.013), the biggest single ablation effect in the whole experiment. Mistral's best performance across all conditions is fine-tuned, temp=0.0, k=3 (ROUGE-1 = 0.092). The mechanism is simple: at k=5, the retrieved chunks plus the prompt frequently exceed the 4,096-token context window, which we confirmed by finding direct overflow errors in the logs.

4.4.3 Latency

Figure 9 shows per-query latency. GPT-4o-mini is the fastest (2.27–2.56s), Claude is slightly slower (2.75–2.82s), and Mistral is slowest (4.64–4.87s). Fine-tuning adds minimal latency overhead for Claude and Mistral, and modest overhead for GPT (+0.29s). The overall mean across conditions is 3.32s. Mistral runs locally on a T4 GPU while GPT and Claude are API calls, so these numbers

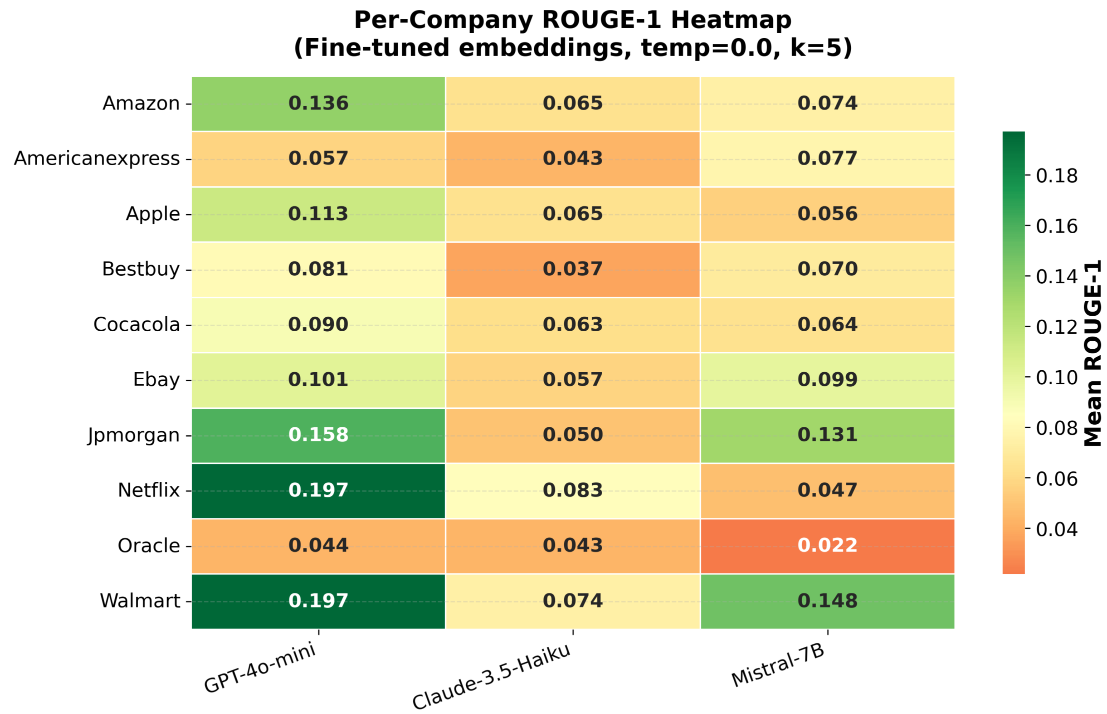

Figure 4: Per-company ROUGE-1 heatmap (fine-tuned embeddings, temp=0.0, k=5). Oracle is the hardest for all three models; Netflix and Walmart are the easiest for GPT-4o-mini.

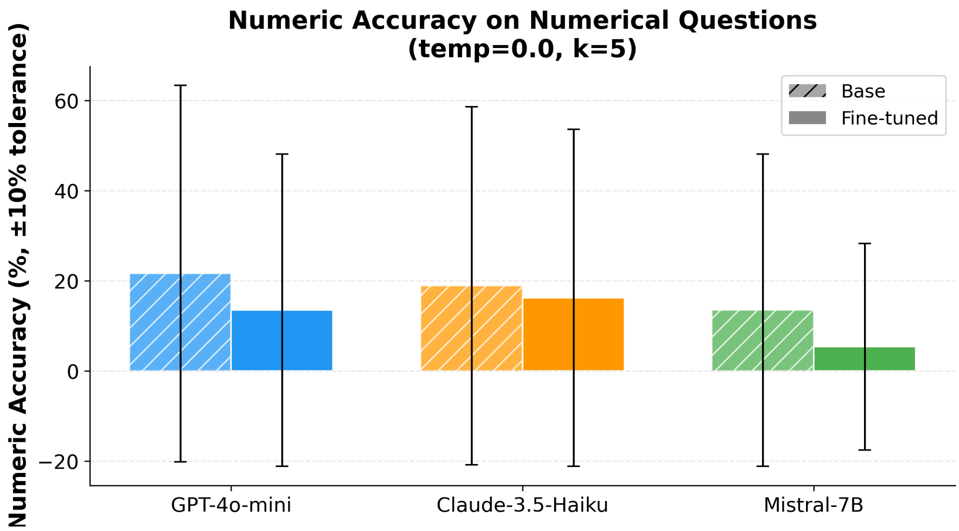

Figure 5: Numeric accuracy on numerical questions ($\pm$10% tolerance, temp=0.0, k=5). Fine-tuning reduces accuracy for all models; GPT-4o-mini and Mistral-7B each drop 8.1 percentage points.

are not directly comparable.

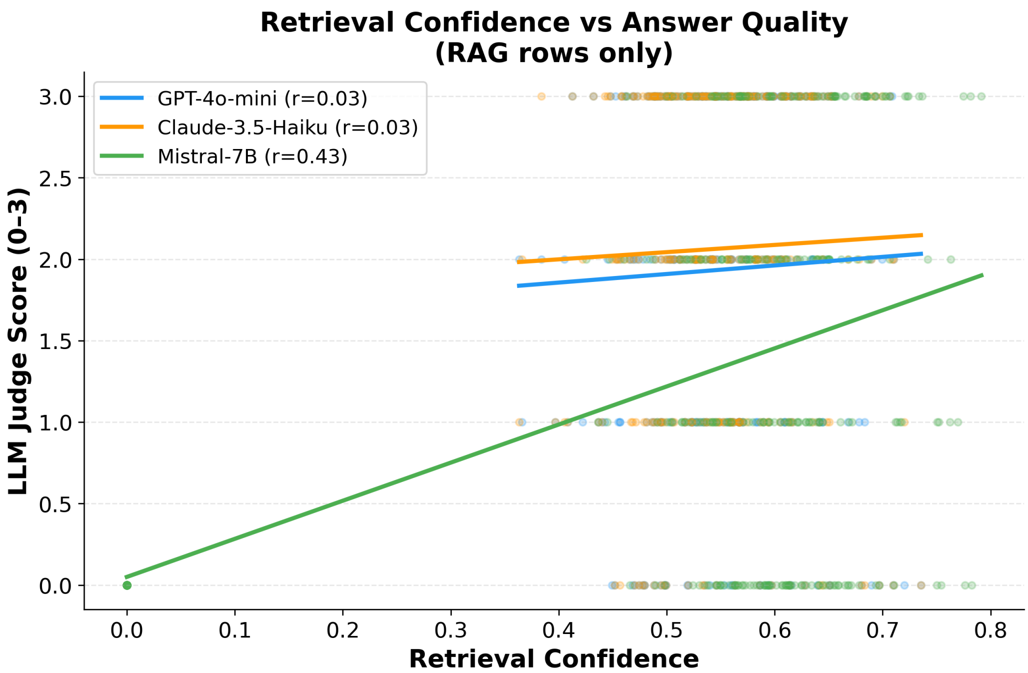

Figure 6: Retrieval confidence vs. LLM Judge score by model (RAG conditions only). GPT-4o-mini and Claude-3.5-Haiku show near-zero correlation (r=0.03, p >0.5); Mistral-7B shows moderate significant correlation (r=0.43, p <0.0001).

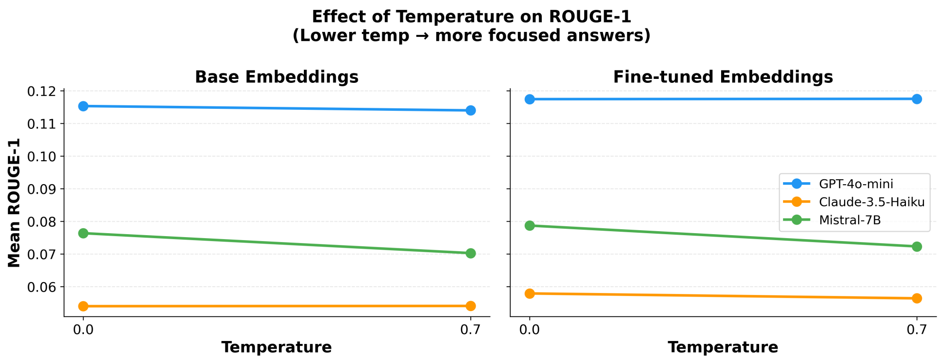

Figure 7: Effect of temperature on ROUGE-1 (k=5, both embedding types). GPT-4o-mini and Claude-3.5-Haiku are largely unaffected; Mistral-7B degrades noticeably at temperature 0.7.

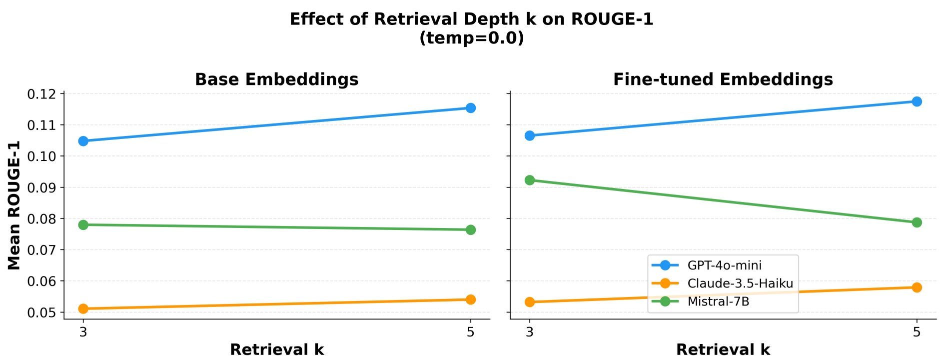

Figure 8: Effect of retrieval depth k on ROUGE-1 (temp=0.0, both embedding types). GPT-4o-mini and Claude-3.5-Haiku improve with k=5; Mistral-7B is hurt due to context window overflow.

Model	Embedding	ROUGE-1 (k=3)	ROUGE-1 (k=5)	Δ (k=3$\rightarrow$5)	Note
GPT-4o-mini	Base	0.105	0.115	+0.010	More context helps
GPT-4o-mini	Fine-tuned	0.107	0.118	+0.011	More context helps
Claude-3.5-Haiku	Base	0.051	0.054	+0.003	More context helps
Claude-3.5-Haiku	Fine-tuned	0.053	0.058	+0.005	More context helps
Mistral-7B	Base	0.078	0.076	−0.002	Context window saturates
Mistral-7B	Fine-tuned	**0.092**	0.079	−0.013	Overflow errors at k=5

Table 3: ROUGE-1 by retrieval depth k (temp=0.0). GPT-4o-mini and Claude-3.5-Haiku consistently improve with more retrieved context. Mistral-7B degrades at k=5 due to context window overflow (4,096 token limit). **Bold** marks Mistral's best performance across all conditions.

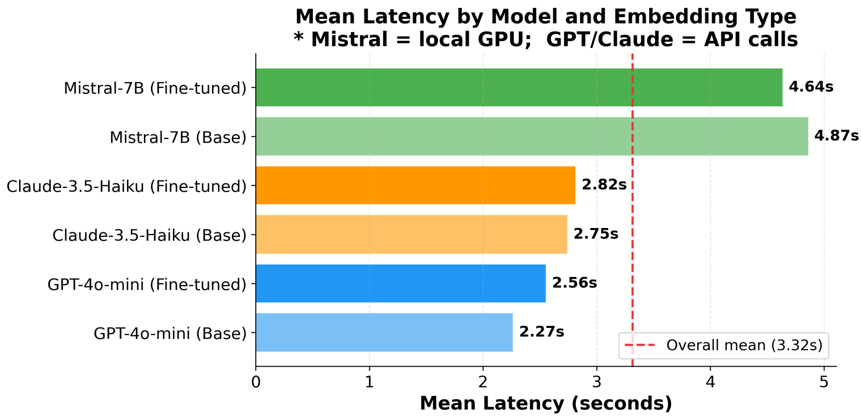

Figure 9: Mean per-query latency by model and embedding type. Mistral-7B runs locally on a Google Colab T4 GPU; GPT-4o-mini and Claude-3.5-Haiku times reflect network API round-trip. The dashed red line marks the overall mean latency of 3.32s.

Model	Fact.	Num.	Tot.	Primary Failure
GPT-4o-mini	10	10	20	Retrieval miss; wrong line item
Claude-3.5-Haiku	6	8	14	Verbose non-answers
Mistral-7B	34	23	57	Context overflow; wrong number

Table 4: Zero-score responses (LLM Judge = 0) by model and question type (fine-tuned, temp=0.0, k=5). Mistral-7B fails at nearly 3× the rate of Claude and 2.5× the rate of GPT. No reasoning failures recorded.

4.5 Case Study: Failure Analysis

We looked at all zero-score responses (LLM Judge = 0) in the fine-tuned, temp=0.0, k=5 condition. Table 4 shows the counts. Mistral fails at nearly 3× the rate of Claude and 2.5× the rate of GPT. No reasoning failures were recorded for any model. Three failure modes account for most of the errors.

Retrieval Miss. For "What was Amazon's advertising services revenue in 2022?" (GT: $38.0B), GPT-4o-mini said the provided excerpts didn't include that figure. That's actually the correct behavior, it didn't hallucinate, but it still scores 0. This type of error would be fixed by better chunking or passage re-ranking, not by tweaking the generation side.

Wrong Number Retrieved. For "What was Amazon's free cash flow in FY2022?" (GT: ≈ −$19.7B), GPT-4o-mini answered $(11,569)M, which is a real Amazon figure but from a different cash flow line item. For "American Express billed business volume?" (GT: $1.5T), the model answered $499.5B, likely a sub-segment number. In both cases the embedding retrieved the right section of the report but not the specific table row needed. This is the main reason we think fine-tuning hurts numeric accuracy: fine-tuned embeddings are better at finding topically relevant passages, but numerical questions often need a specific row in a specific table, and keyword overlap from base embeddings sometimes gets you there more reliably. Future work should look at table-aware chunking and structured number extraction to address this properly.

Context Window Overflow. This only happened with Mistral-7B. For "What was AWS operating income in FY2022?" (GT: $22.8B), the model returned: *"Requested tokens (4,378) exceed context window of 4,096."* The model never even saw the retrieved content. This is fully preventable by using k=3 for context-limited models.

Claude's Verbose Non-Answers. Claude's failures were a different kind of problem: structured, well-formatted responses with headers that acknowledged the question but didn't actually extract the right value. Claude generates high-quality-looking outputs even when the retrieved context isn't sufficient, which explains both its strong LLM Judge scores overall and its non-zero numerical failure rate.

5 Conclusion

Fine-tuning consistently improves ROUGE-1 (+0.002 to +0.004) and Mistral's judge score (+0.060), but the biggest factor in answer quality is whether you have retrieval at all. Adding RAG improves Claude's judge score by +0.834; switching to fine-tuned embeddings on top of that adds +0.000. Fine-tuning also helps in a question-type-dependent way: it helps factual and reasoning questions but hurts numeric accuracy across all three models, likely because it optimizes for topical relevance over precise numeric retrieval. Financial numerical QA is still largely unsolved: accuracy peaks at 21.6% and drops with fine-tuning. Model-specific configuration also matters: Mistral needs k=3 and temp=0.0 or the context window becomes a hard constraint; commercial models perform best at k=5 and temp=0.0. Finally, ROUGE is a weak metric for this kind of task and should always be paired with a semantic metric, reporting ROUGE alone will systematically disadvantage verbose but accurate models like Claude.

Limitations

The fine-tuning dataset is only 35 pairs, which is small, and results could shift with more training data. We only evaluate on 2022 10-K filings from ten companies, so generalization to other document types or time periods is an open question. LLM-as-judge scores are produced by GPT-4o-mini, which introduces potential self-evaluation bias. Mistral ran in 4-bit quantization, which likely reduces quality compared to full precision.

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